Infrastructure as Capacity Ceiling: The Physical Preconditions of Consolidation

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November 3, 2026

Abstract

The consolidation theory treats capacity K as a choice variable: the operator picks a vehicle size to lower the break-even load factor $\lambda^* = c_0/[K(p - c_v)]$. This paper argues that capacity is not freely chosen. It is bounded by **infrastructure**, and the bound is the physical precondition for consolidation. We formalise infrastructure quality I as a parameter that enters the theory through three distinct channels: it **caps the feasible capacity** $\mathcal{K}(I)$, which sets a floor on λ^* that no management can lower; it **raises variable cost** $c_v(I)$, which raises λ^* by a second independent channel; and it **inflates travel-time variance** $\mathrm{Var}[\tau(I)]$, which destroys the schedulability condition that the premium product depends on. The three channels compound. We then show that infrastructure generates the same coordination failure as addressability: bad infrastructure caps capacity, which caps volume, which removes the justification to invest in infrastructure. The result is two stable equilibria separated by a separatrix, structurally identical to the addressability death spiral. We derive the **substitution boundary** between digital and physical infrastructure: digital can substitute for the *information* layer (addressing, coordination, payments) but never for the *mobility* layer. This gives the portfolio rule: digital investment has the highest return where mobility infrastructure already exists and only the information layer is missing. The paper's central claim is that **the mode observed in a corridor is the equilibrium of its infrastructure, not a stage of its development**, and that the boda-boda is not Africa's transport problem but the equilibrium mode of a corridor whose infrastructure admits nothing larger.

1 The Puzzle

The consolidation theory predicts which mode should dominate a corridor given its demand density. It does not explain why the same demand density produces different modes in different places. A dense corridor in Nairobi is served by matatus; a dense corridor of similar population in London is served by buses; a dense corridor in Tokyo is served by trains.

The conventional explanation invokes income and preference: richer places buy larger vehicles, poorer places buy smaller ones. This is partly true and mostly incomplete. A matatu operator in Nairobi would happily buy a bus if the bus could complete its route. A bus operator in London would happily buy a train if the train could be justified. The choice is not a preference. It is a constraint.

The constraint is **infrastructure**. This paper argues that infrastructure is not a background condition of logistics. It is a **capacity ceiling**, and the ceiling determines the equilibrium mode. The mode observed in a corridor is not a stage of development. It is the solution to a specific physical problem: how large a vehicle can this corridor admit.

2 Infrastructure as a Parameter

Definition 1 (Infrastructure quality)**.** *Let $I \in (0, 1]$ denote infrastructure quality, where $I = 1$ represents a corridor with full formal infrastructure (paved roads of adequate width, bridges of*

adequate clearance, addressing, drainage, signage) and $I \to 0$ represents a corridor with no formal infrastructure at all.

Infrastructure enters the theory through three distinct channels, each of which raises the break-even load factor.

2.1 Channel 1: the feasible capacity set

Definition 2 (Feasible capacity set)**.** *For a corridor with infrastructure quality I, the feasible capacity set is*

$$\mathcal{K}(I) = \{K : K \text{ can physically complete the corridor at infrastructure } I\}. \tag{1}$$

Let $K_{\max}(I) = \max \mathcal{K}(I)$ be the maximum feasible capacity.

Proposition 1 (Capacity Ceiling Proposition)**.** *$K_{\max}(I)$ is increasing in I, and there exists a threshold I^* such that for $I < I^*$, $K_{\max}(I) = 1$: only a motorcycle can complete the corridor.*

The threshold I^* corresponds to the physical minimum: a path narrow enough or steep enough that no four-wheeled vehicle can pass. In an informal settlement with unpaved narrow paths, $I < I^*$ and the only feasible mode is the boda-boda. In a paved urban corridor, $I \gg I^*$ and the feasible capacity is bounded only by demand.

2.2 Channel 2: variable cost

Definition 3 (Variable cost function)**.** *Variable cost per unit carried is*

$$c_v(I) = c_v^0 + \gamma \cdot (1 - I), \tag{2}$$

where c_v^0 is the cost on a perfect corridor and γ is the damage coefficient (fuel, tyres, suspension, maintenance). $c_v(I)$ is decreasing in I.

Potholes, unpaved surfaces, and poor drainage raise the cost of moving a unit. They do not raise it uniformly — they raise it disproportionately for heavier vehicles, because the damage caused by an axle scales roughly with the fourth power of the load. So the damage coefficient γ is itself increasing in K. This will matter in Section 4.

2.3 Channel 3: travel-time variance

Definition 4 (Variance function)**.** *The variance of travel time on a corridor of infrastructure I is*

$$\mathrm{Var}[\tau_{travel}(I)] = \sigma^2_{travel}(I), \tag{3}$$

which is decreasing in I. On unpaved roads, weather, potholes, and unpredictable traffic raise the variance above what the timetable can absorb.

This third channel is the one that matters most commercially, and it is the one that is usually ignored. A corridor with bad roads does not merely cost more to traverse. It **cannot be scheduled**. And a logistics business that cannot be scheduled cannot sell the product that differentiates it from the informal market.

3 The Double Hit on the Threshold

The break-even load factor under infrastructure I is

$$\lambda^*(I) = \frac{c_0}{K_{\max}(I) \cdot \big[p - c_v(I)\big]}. \tag{4}$$

Proposition 2 (The Double Hit). *$\lambda^*(I)$ is decreasing in I through* two independent channels*:*

1. *As I falls, $K_{\max}(I)$ falls, which raises λ^* by reducing the capacity over which the fixed cost is amortised.*
2. *As I falls, $c_v(I)$ rises, which raises λ^* by reducing the net margin per unit.*

The two effects multiply, not add. The threshold under poor infrastructure is higher than the sum of the two effects considered separately.

Proof. Both effects operate on the denominator of λ^*. Write the denominator as $K_{\max}(I) \cdot [p - c_v(I)]$. A fall in $K_{\max}$ and a rise in c_v both reduce this product. The reduction is multiplicative because both are factors of the same product. The second-order cross-term is positive (both effects push in the same direction), so the combined effect exceeds the sum. □

Corollary 1 (The infrastructure trap on capacity). *A corridor with poor infrastructure faces a threshold that is higher than the corridor's demand can clear, not because the demand is insufficient but because the infrastructure caps the capacity through which the demand could be served. The boda-boda is the* equilibrium mode *of such a corridor, not a sign of underdevelopment.*

Infrastructure I	$K_{\max}$	c_v	λ^*	Dominant mode
Formal urban ($I \approx 1$)	1,300+	Low	0.33	Train
Paved arterial ($I \approx 0.8$)	60	Low	0.93	Bus
Paved minor ($I \approx 0.6$)	40	Medium	0.70	Van
Unpaved road ($I \approx 0.3$)	14	High	0.93	Matatu
Informal path ($I \approx 0.1$)	1	Very high	degenerate	Boda-boda

The table is not a development ladder. It is an **equilibrium table**: each row is the mode a rational operator would choose given the infrastructure available. Nothing is "behind." Everything is locally optimal given I.

4 The Third Channel: The Death of the Premium Product

The first two channels make delivery expensive. The third channel makes it **unschedulable**. And unschedulability is a more fundamental problem, because it destroys the product itself.

Recall the schedulability condition from the addressability theory: a service window of width $\pm W$ is promisable only if

$$\mathrm{Var}[\tau_{\text{total}}] \leq \Psi(W), \tag{5}$$

where

$$\mathrm{Var}[\tau_{\text{total}}] = \mathrm{Var}[\tau_{\text{service}}] + \mathrm{Var}[\tau_{\text{travel}}(I)]. \tag{6}$$

Proposition 3 (The Schedulability Floor). *There exists an infrastructure level I_S below which* ***no commercially useful window can be promised at any price****. Formally,*

$$I < I_S \implies \mathrm{Var}[\tau_{travel}(I)] > \Psi(W) \quad \textit{for all } W \in [W_{\min}, W_{\max}]. \tag{7}$$

Corollary 2 (No premium product in low-infrastructure corridors)**.** *In a corridor with $I < I_S$, the operator cannot sell a guaranteed delivery window, because the physics will not support the guarantee. The premium tier does not merely become expensive — it* ***cannot exist****. The customer cannot buy reliability at any price, because reliability is not physically producible in that corridor.*

This is the deepest commercial consequence of infrastructure. It is not that bad roads make delivery expensive; it is that bad roads make the *guarantee* impossible, and the guarantee is the product. Without schedulability, the operator is not in the premium business. He is in the spot business, and the spot business has the margins of the spot business.

Proposition 4 (The two-product proposition)**.** *The product mix available in a corridor is determined by its infrastructure:*

1. *$I > I_S$: both spot and premium products are available. The operator can choose the mix and price the premium at a markup.*
2. *$I \leq I_S$: only the spot product is available. The premium tier cannot be offered, and the operator's margin is bounded by the spot margin.*

This is why the formal logistics business cannot establish itself in low-infrastructure corridors, and why the informal market dominates there. It is not a question of capability or capital. It is a question of physics. The physics of the road determine whether the guarantee is possible, and the guarantee is the product.

5 The Infrastructure Gap

The infrastructure caps the feasible capacity. The demand density determines the optimal capacity. The gap between the two is the infrastructure gap.

Definition 5 (Infrastructure gap)**.** *For a corridor with demand density D and infrastructure I, the infrastructure gap is*

$$\Gamma(D, I) = K^*(D) - K_{\max}(I), \tag{8}$$

where $K^*(D)$ is the capacity that would minimise λ^* given the demand density.

Proposition 5 (The Infrastructure Gap Proposition)**.** *If $\Gamma > 0$, the corridor is* ***infrastructure-constrained****: it cannot achieve its economically optimal capacity, and λ^* is higher than it would be with adequate infrastructure. The loss from the gap is*

$$\mathcal{L}(D, I) = \lambda^*(I) - \lambda^*(I_{adequate}) > 0. \tag{9}$$

The infrastructure gap has three properties that the viability gap of Document 11 does not have.

First, it is large. The viability gap is measured in units of overload — a few passengers or tonnes. The infrastructure gap is measured in *orders of magnitude* — the difference between a motorcycle and a train is a factor of a thousand in capacity.

Second, it is not closeable by the operator. The viability gap can be closed by overloading. The infrastructure gap cannot, because the road is not the operator's to fix.

Third, it is self-perpetuating. The gap suppresses volume, and low volume removes the justification to close the gap. This is the infrastructure trap, and it is the subject of the next section.

6 The Infrastructure Trap

The infrastructure gap has the same structure as the addressability death spiral.

Proposition 6 (The Infrastructure Trap)**.** *The corridor has two stable equilibria:*

1. ***Low-infrastructure equilibrium****: poor roads cap K, which raises λ^*, which suppresses volume, which removes the justification to invest in the roads. The corridor stays at $I \approx 0.1$ and the boda-boda remains the equilibrium mode.*

2. ***High-infrastructure equilibrium****: adequate roads allow $K \gg 1$, which lowers λ^*, which supports volume, which justifies further investment. The corridor converges to $I \approx 1$ and the train or bus is the equilibrium mode.*

The two equilibria are separated by a separatrix I_{sep}, and a corridor at $I < I_{sep}$ has no incentive to invest its way out.

Sketch. The volume in a corridor is a function of λ^* and of demand density. The investment in infrastructure is a function of volume. At low I, λ^* is high, volume is low, and the expected return on infrastructure investment is below its cost. At high I, λ^* is low, volume is high, and the expected return exceeds cost. The two are self-consistent; the intermediate case is unstable. □

Corollary 3 (No private actor will close the gap)**.** *Infrastructure is **non-excludable** (no one can be prevented from using a road) and **non-rival at the margin** (one more vehicle does not degrade a well-built road). No private actor can capture the return on infrastructure investment, so no private actor will make it. The gap can only be closed by the state or by a consortium acting with state-like powers.*

This is the same argument as for the address layer, and the conclusion is the same: **the institutional supply of the missing asset must come from outside the market**. The state supplies the road; the platform supplies the ledger. Neither can be supplied by the other.

7 The Substitution Boundary

This section states the honest boundary of the entire series.

Digital infrastructure can substitute for some missing physical infrastructure. It cannot substitute for all of it. The line between the two is precise.

Definition 6 (The information-mobility decomposition)**.** *A logistics system has two physical layers:*

- ***The information layer****: addressing, coordination, dispatch, payments, tracking, proof of delivery.*

- ***The mobility layer****: the physical movement of goods from origin to destination.*

Proposition 7 (The Substitution Boundary)**.** *Digital infrastructure can substitute for the **information layer** but not for the **mobility layer**:*

1. ***Addressing****: a digital ledger can record geocodes, landmarks, and access points even where no formal addressing exists. The ledger substitutes for the missing address system.*

2. ***Coordination****: a platform can dispatch, pool, and coordinate even where no formal transport authority exists. The platform substitutes for the missing coordination system.*

3. ***Payments****: mobile money can settle transactions even where no formal banking exists. The payment layer substitutes for the missing banking system.*

4. ***Mobility: nothing digital can make a road passable.*** *A ledger cannot move a parcel across a river, up a hill, or through a flood. The mobility layer is irreducibly physical.*

Corollary 4 (The leapfrog is real but partial)**.** *East Africa leapfrogged branch banking with mobile money because* ***payments required no physical infrastructure****. It cannot leapfrog roads, because* ***a parcel still has to physically move****. The leapfrog is real for the information layer and impossible for the mobility layer.*

Corollary 5 (The digital-investment rule)**.** *Digital investment has the highest return in corridors where the mobility infrastructure already exists and only the information layer is missing. In corridors where mobility is also missing, digital investment is necessary but insufficient: it raises the achievable rate $R(a)$ but does not raise the feasible capacity $K_{\max}(I)$.*

This is the portfolio rule that the entire series has been building toward. **The Peleka thesis is a bet on the information layer**, and the bet is correct in every corridor where the mobility layer is already there — which is to say, in dense urban corridors. In corridors where the mobility layer is also missing, the digital layer is the right first investment but it is not a complete solution, and the corridor will not reach the high-density equilibrium until the mobility layer is also supplied.

Missing layer	Digital substitute?	Private supply?	Correct response
Addressing	**Yes**	Yes (ledger)	Build the ledger
Coordination	**Yes**	Yes (platform)	Build the platform
Payments	**Yes**	Yes (mobile money)	Integrate mobile money
Mobility	**No**	No (non-excludable)	State must supply

8 The Geography of the Theory

The theory now predicts the *geography* of logistics, not only its dynamics.

Proposition 8 (The Geography Proposition)**.** *The dominant mode in a corridor is the equilibrium of its infrastructure, not a stage of its development:*

1. ***Low*** *I: $K_{\max} = 1$, the boda-boda is the equilibrium mode, and the residual demand is the whole market.*
2. ***Medium*** *I: $K_{\max} = 14$, the matatu is the equilibrium mode, and the residual is large but not universal.*
3. ***High*** *I: $K_{\max} \geq 60$, the bus or train is the equilibrium mode, and the residual is small.*

Every corridor settles on the mode its infrastructure can support. Nothing is "behind." Everything is locally optimal given I.

This is the answer to the boda-boda question. The boda-boda is not a feature of African culture or Asian development. It is the **equilibrium mode of a corridor whose infrastructure admits nothing larger**. In corridors where $I > I^*$, it is dominated on every dimension except speed, and it survives only as a niche service for time-critical drops that cannot wait for a larger vehicle.

Corridor type	I	$K_{\max}$	Dominant mode	Residual
Informal settlement	0.1	1	Boda-boda	Everything
Rural road	0.3	14	Matatu	Large
Urban arterial	0.8	60	Bus	Small
Rail corridor	0.95	1,300	Train	Very small

Institution	Type	Supplier	Why
Mobility infrastructure	Public good	State	Non-excludable; no private capture of returns
Aggregation points (stages, ports, hubs)	Public good	State or common	Non-rival, non-excludable; free-rideable
Digital ledger (platform, addressing, dispatch)	Club good	Private	Excludable, costly to build, returns capturable

9 The Institutional Allocation

The theory now yields a complete allocation of consolidation institutions to suppliers.

This allocation is not a policy preference. It is a consequence of the **excludability** of each institution:

- **The road cannot be sold per use** without a tolling apparatus that costs more than the road. It is non-excludable, so the market cannot supply it.
- **The stage cannot be sold per use** without a gate that costs more than the stage. It is non-excludable, so the market cannot supply it.
- **The platform can be sold per use** because the app is a gate. It is excludable, so the market can supply it.

The boda-boda industry has converged on using all three: free-riding on stages, riding on roads, paying for platforms. That is the correct institutional allocation for its demand type, and it emerged without a planner because the economics of excludability determined it.

10 Policy Implications

The infrastructure analysis yields five policy rules.

1. **Do not set vehicle standards without reference to infrastructure.** A standard that requires buses on roads that cannot admit them is a standard that produces no buses.
2. **Close the infrastructure gap before formalising the mode.** Formalising the boda-boda without upgrading the road locks the corridor into the low-infrastructure equilibrium, because formalisation removes the flexibility that the informal mode relies on.
3. **Invest in the information layer first where the mobility layer exists.** Digital investment has the highest return in corridors where the road is already there and the ledger is not.
4. **Treat infrastructure investment and logistics investment as complements.** Neither alone can close the gap. The state supplies the road; the private sector supplies the ledger; the two must be coordinated in time.
5. **Measure the schedulability floor.** A corridor with $I < I_S$ cannot support the premium product, and no amount of private investment in the premium product will succeed. This is a physical constraint, and it must be measured before it is planned around.

11 Falsifiability

Six observations would refute the theory.

1. **Against the capacity ceiling**: evidence that a mode larger than $K_{\max}(I)$ operates successfully on a corridor with infrastructure I.
2. **Against the double hit**: evidence that λ^* is lower on poor infrastructure than on good, holding demand constant.
3. **Against the schedulability floor**: evidence that a premium product with a guaranteed window succeeds on a corridor with $\mathrm{Var}[\tau(I)] > \Psi(W)$.
4. **Against the infrastructure trap**: evidence that a corridor at $I < I_{\text{sep}}$ escapes to the high-infrastructure equilibrium without external investment.
5. **Against the substitution boundary**: evidence that digital infrastructure makes a physically impassable corridor passable.
6. **Against the geography proposition**: evidence that the same infrastructure supports different dominant modes in different places.

Each is testable in any corridor to which the theory is applied.

12 Conclusion

Infrastructure is not a background condition. It is a capacity ceiling, and the ceiling determines the equilibrium mode.

Bad infrastructure caps the feasible capacity, which sets a floor under the break-even load factor that no management can lower. It raises variable cost by a second independent channel, so the two effects compound. And it inflates travel-time variance, which destroys the schedulability condition that the premium product depends on. The first two channels make delivery expensive; the third makes the guarantee impossible, and the guarantee is the product.

The infrastructure trap is structurally identical to the addressability death spiral: bad infrastructure caps capacity, which caps volume, which removes the justification to invest in infrastructure. The two equilibria are separated by a separatrix, and no private actor will close the gap, because infrastructure is non-excludable and the returns cannot be captured.

The substitution boundary is precise. Digital infrastructure can substitute for the missing information layer — addressing, coordination, payments. It cannot substitute for the missing mobility layer. The leapfrog is real for the information layer and impossible for the mobility layer. This gives the portfolio rule: digital investment has the highest return where the road is already there and the ledger is not.

And the geography of logistics follows. The mode observed in a corridor is the equilibrium of its infrastructure, not a stage of its development. The boda-boda is not Africa's transport problem. It is the equilibrium mode of a corridor whose infrastructure admits nothing larger.

> **The theory in one sentence:** infrastructure is a capacity ceiling that sets a floor on the break-even load factor no management can lower, raises costs by a second channel, destroys the schedulability the premium product depends on, and generates the same coordination trap as addressability — so the mode observed in a corridor is the equilibrium of its infrastructure, not a stage of its development, and digital investment can substitute for the missing information layer but never for the missing road.